

Real Estate Support Triage Agent Report

Course Name: Agentic AI

Institution Name: Medicaps University – Datagami Skill Based Course

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Academic Year: 2026

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1. Problem Statement & Objectives

1.1 Problem Statement

In the real estate and property management domain, handling tenant queries manually is time-consuming, inconsistent, and inefficient. Property managers receive a large number of unstructured messages related to maintenance, complaints, payments, and inquiries.

The key challenges include:

- Delay in responding to tenants
- Lack of prioritization (urgent vs non-urgent issues)
- Manual effort in understanding and replying
- Inconsistent communication quality

To solve this, an intelligent automated system is required that can:

- Understand natural language messages
- Identify urgency and intent
- Extract important details
- Generate professional responses automatically

1.2 Project Objectives

The main objectives of this project are:

- To automate tenant communication handling using AI
- To classify incoming messages based on urgency and intent
- To extract structured data from unstructured text
- To generate context-aware professional replies
- To build a scalable and modular agent-based system
- To reduce human effort and improve response time

1.3 Scope of the Project

This project focuses on designing and implementing an AI-powered triage system that:

- Processes tenant messages end-to-end
- Uses an agent-based pipeline (classification → extraction → response)
- Provides REST API-based interaction
- Generates structured JSON outputs

Out of Scope:

- Deployment infrastructure (CI/CD)
- Advanced analytics dashboards
- Full frontend production UI

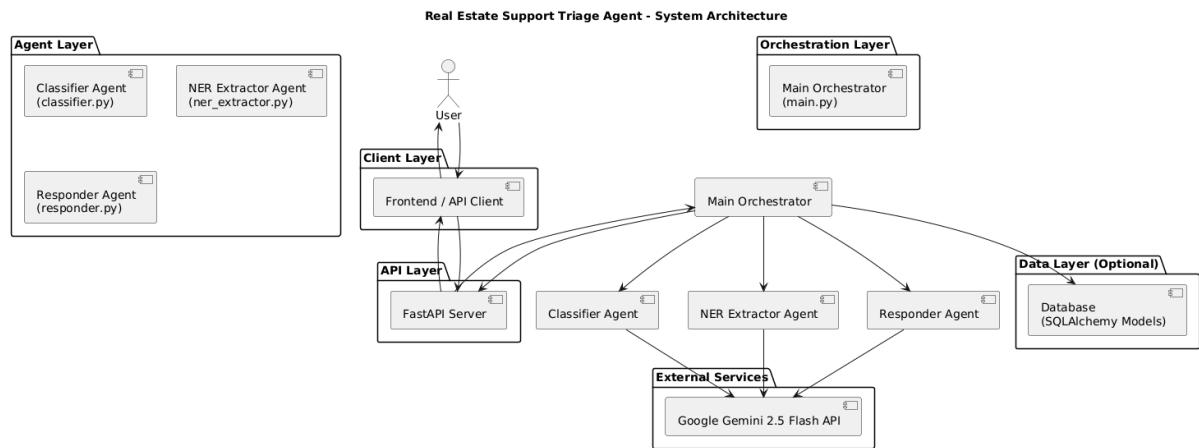
2. Proposed Solution

2.1 Key Features

The system provides the following key features:

- **Message Classification**
Detects urgency (HIGH / MEDIUM / LOW) and intent (complaint, inquiry, etc.)
- **Named Entity Recognition (NER)**
Extracts:
 - Tenant ID
 - Property details
 - Issue type
 - Dates
- **Automated Response Generation**
Creates professional and context-aware replies
- **End-to-End Automation**
No human intervention required
- **REST API Integration**
Easy integration with frontend or external systems
- **Structured JSON Output**
Ensures consistency and usability

2.2 Overall Architecture / Workflow



The system follows a **multi-agent pipeline architecture**:

Workflow Steps:

1. **Message Input**
User sends a request via /triage API
2. **Classification Agent**
 - Identifies urgency
 - Determines intent
3. **NER Extraction Agent**
 - Extracts structured data from text
4. **Response Generation Agent**
 - Generates a professional reply
5. **Final Output**
 - Returns structured JSON response

Architecture Layers:

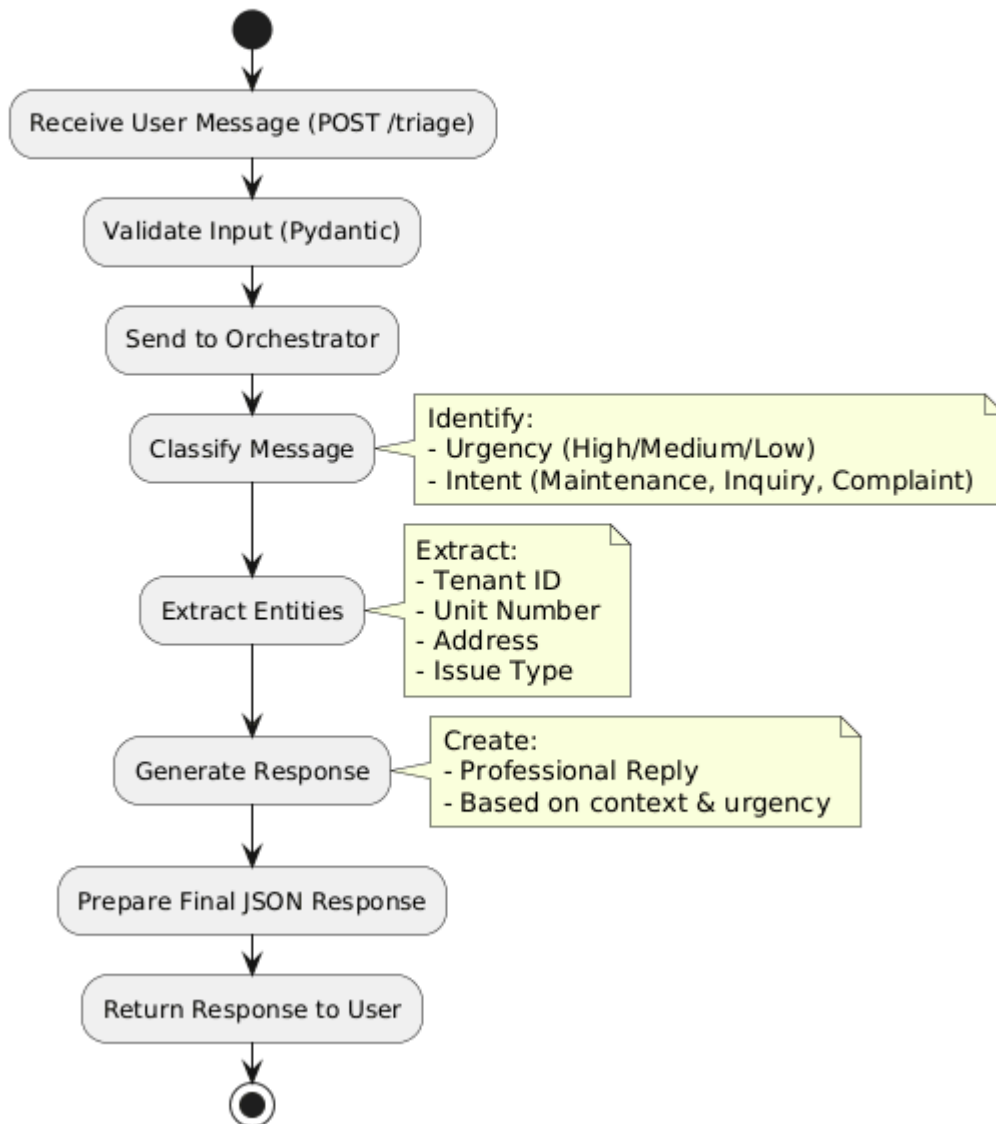
- **API Layer (FastAPI)**
Handles requests and responses
- **Orchestration Layer (main.py)**
Controls flow between agents

- **Agent Layer**
 - Classifier
 - NER Extractor
 - Responder
- **Data Layer**
 - Stores triage information (planned using ORM)

UML Diagrams

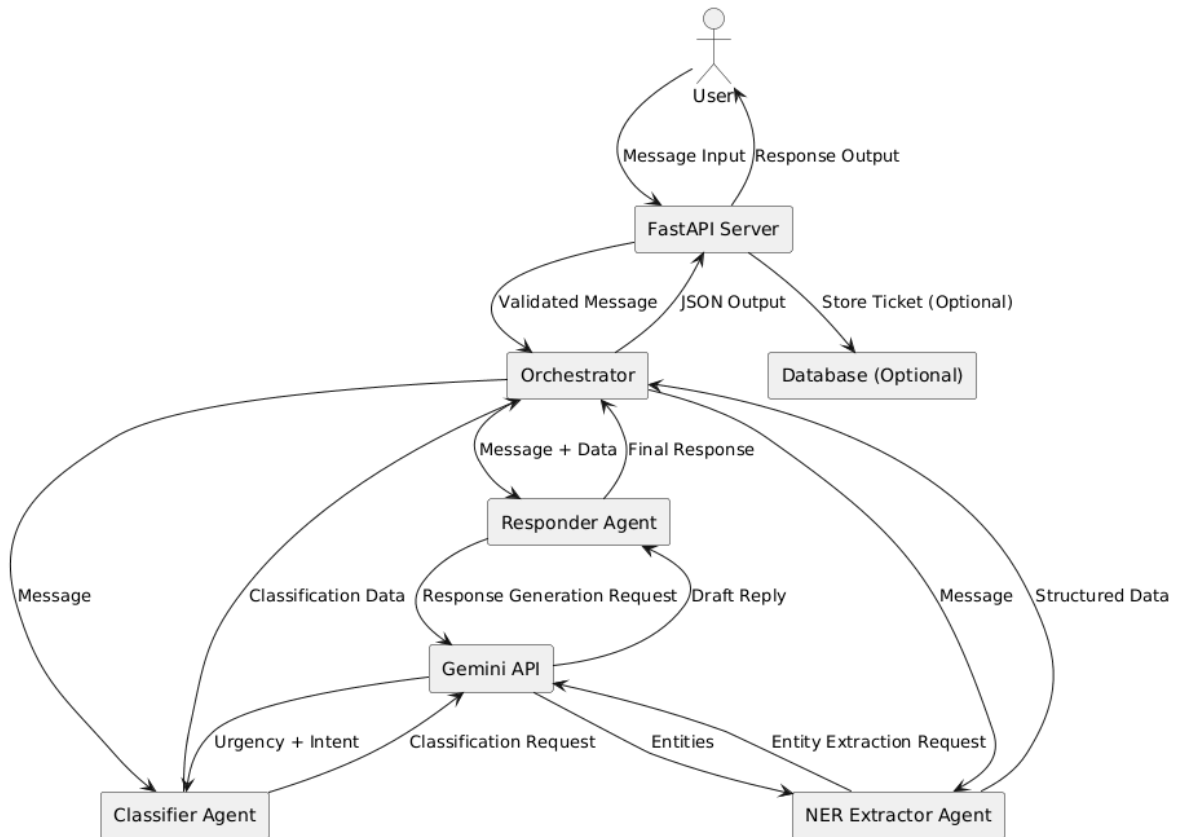
1. Process Flow Diagram

Real Estate Support Triage Agent - Process Flow



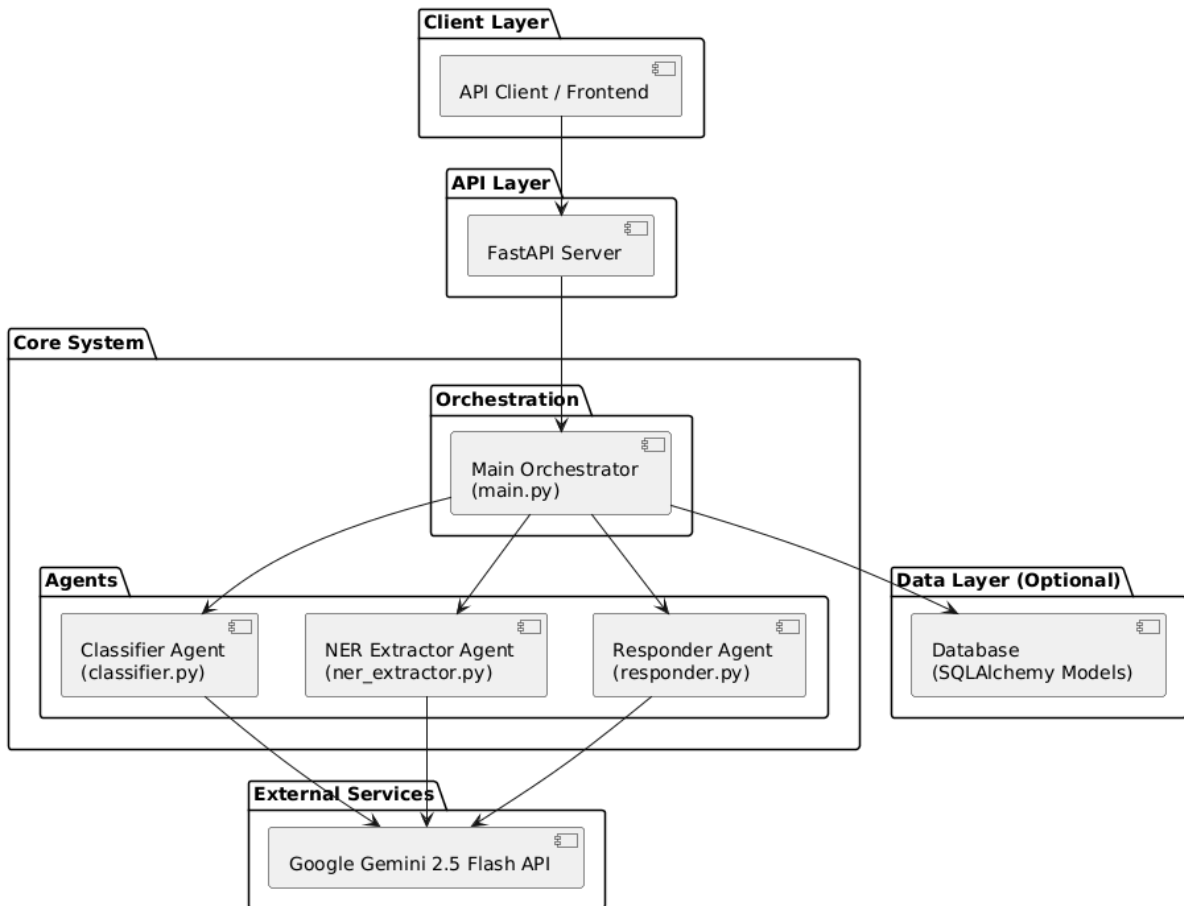
2. Data Flow Diagram

Real Estate Support Triage Agent - Data Flow Diagram

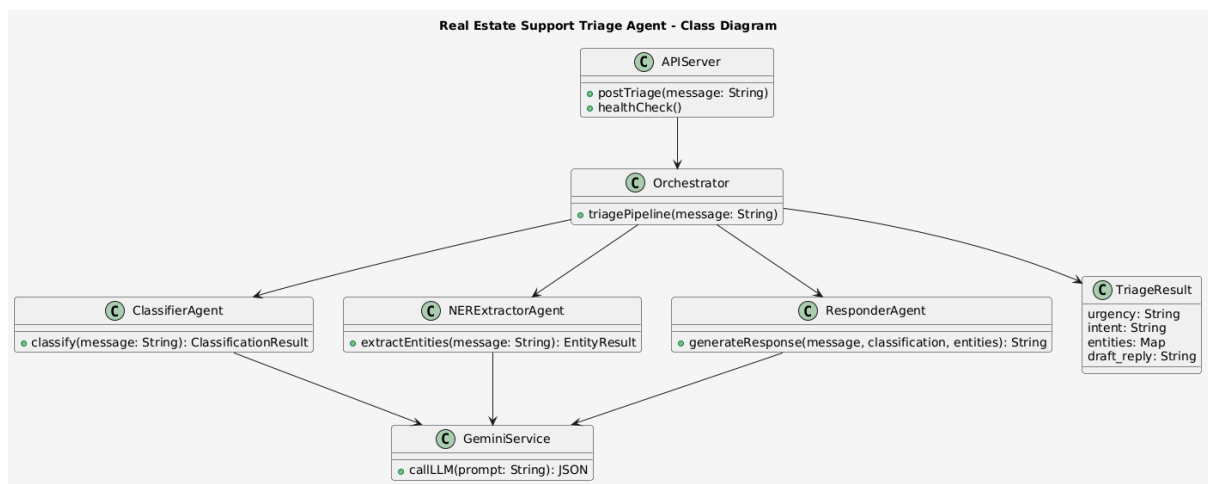


3. Component Diagram

Real Estate Support Triage Agent - Component Diagram

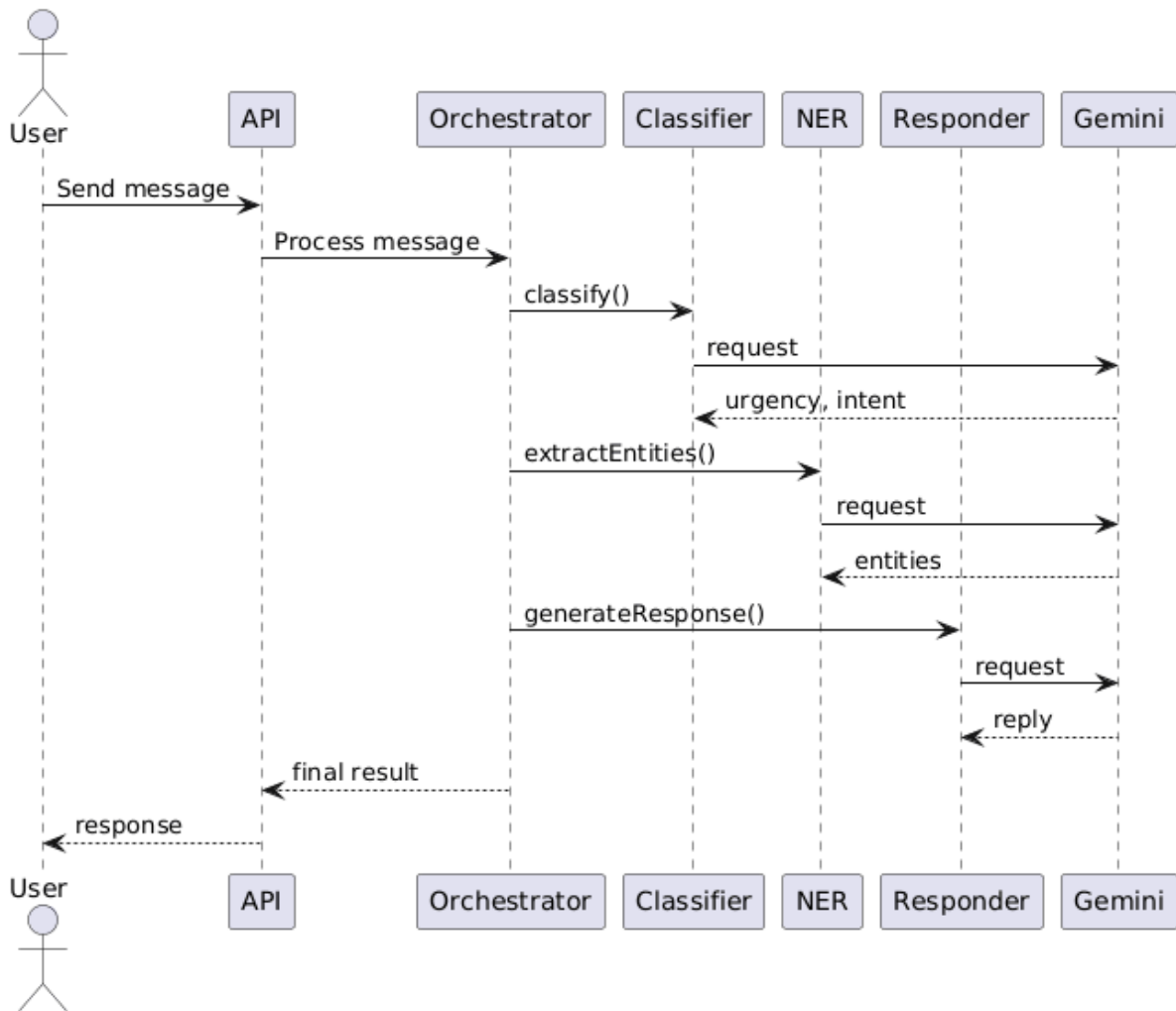


4. Class Diagram



5. Sequence Diagram

Real Estate Support Triage Agent - Simple Sequence Diagram



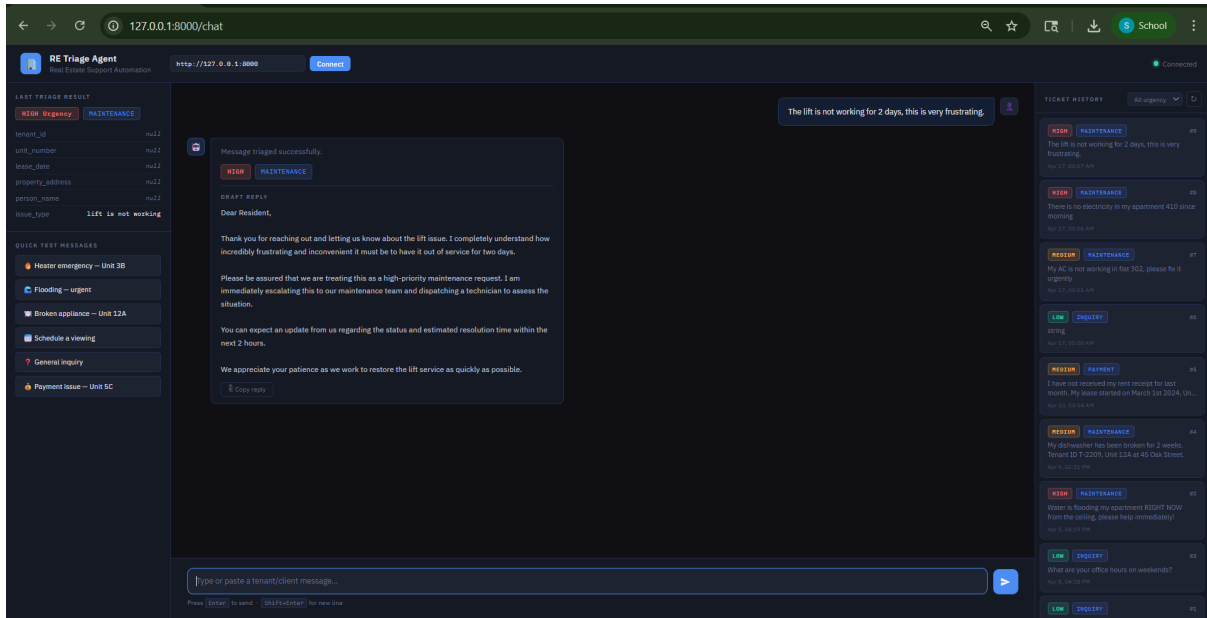
2.3 Tools & Technologies Used

Category	Technology
Programming Language	Python
Backend Framework	FastAPI
AI Model	Google Gemini 2.5 Flash
AI Framework	LangChain
Data Validation	Pydantic
Database	SQLAlchemy (SQLite)
Server	Uvicorn
API Testing	Swagger UI

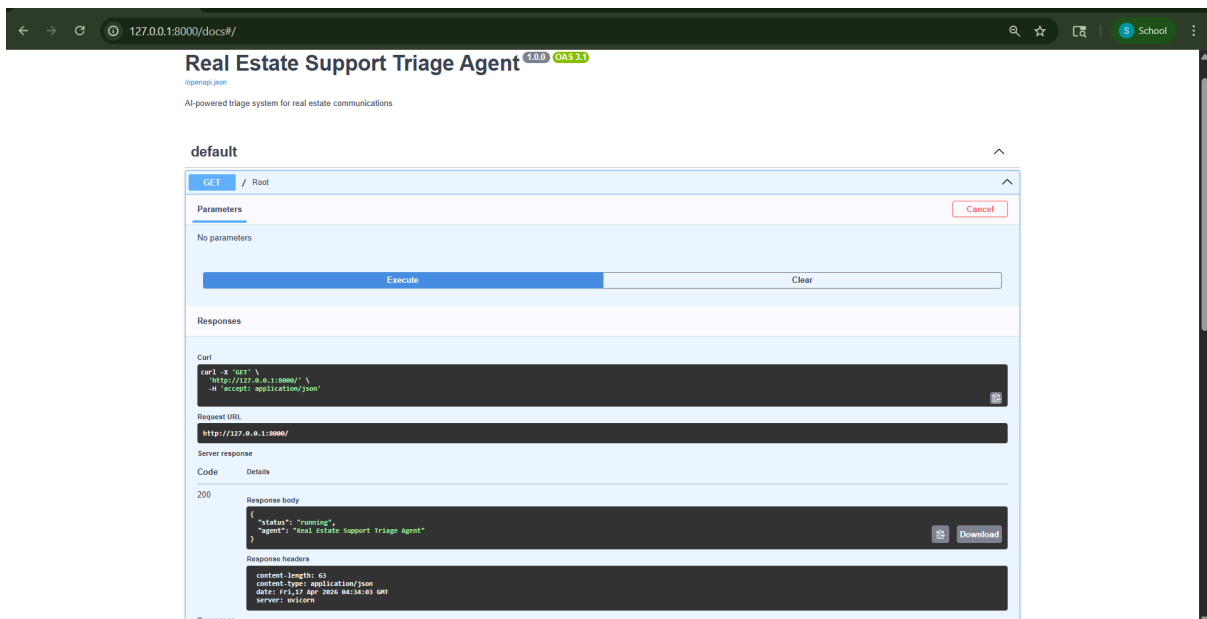
3. Results & Output

3.1 Screenshots / Outputs

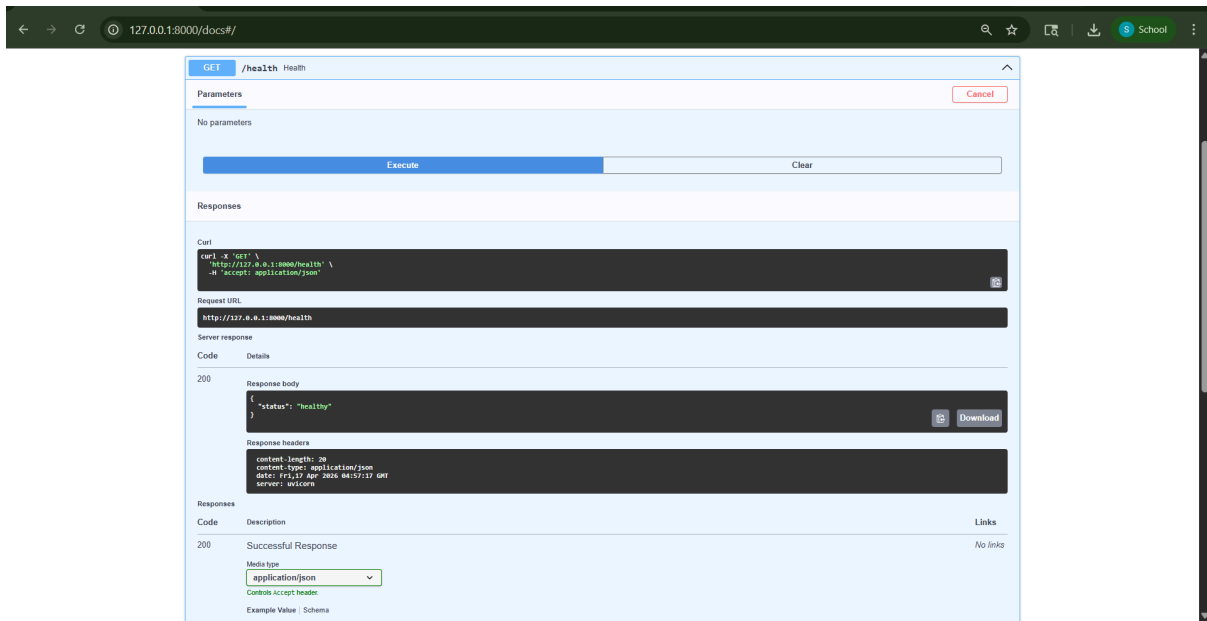
1. Chat Interface / Frontend UI Screenshot



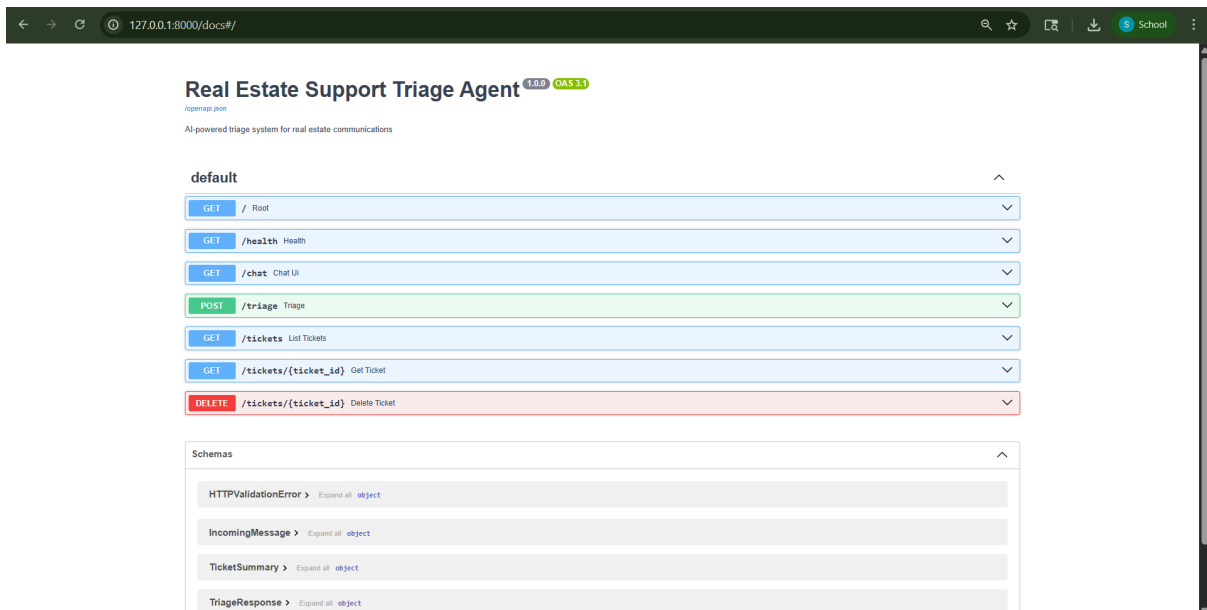
2. Home / Root Endpoint



3. Health Check Endpoint



4. Swagger UI (API Documentation Page)



5. Triage API Request (Input)

POST /trriage Triage

Parameters

No parameters

Request body required

application/json

Edit Value | Schema

```
{
  "message": "My AC is not working in flat 302, please fix it urgently"
}
```

Execute

Clear

6. Triage API Response (Output)

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/trriage' \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "message": "My AC is not working in flat 302, please fix it urgently"
  }'
```

Request URL

```
http://127.0.0.1:8000/trriage
```

Server response

Code

Details

200

Response body

```
{
  "id": 1,
  "original_message": "My AC is not working in flat 302, please fix it urgently",
  "urgency": "MEDIUM",
  "intent": "maintenance",
  "entities": {
    "tenant_id": null,
    "unit_number": "302",
    "lease_date": null,
    "property_address": null,
    "person_name": null,
    "dates": null,
    "issue_type": "AC is not working"
  },
  "draft_reply": "Subject: Regarding Your AC Issue in Unit 302\nDear Resident,\nThank you for reaching out to us. We're very sorry to hear that your AC is not working in unit 302, and we understand how uncomfortable that must be, especially with the weather.\nPlease be assured that we're treating this with the urgency it deserves. We are arranging for a maintenance technician to inspect and repair your AC unit. You can expect action on this within the next 24 hours.\nWe will keep you updated regarding the technician's schedule and estimated arrival time.\nThank you for your patience and understanding.\nWarmly,\n[Your Name/Property Management Team]",
  "source": "web",
  "created_at": "2026-04-17T05:01:41.180659"
}
```

Download

Response headers

```
access-control-allow-origin: *
content-length: 1010
content-type: application/json
date: Fri, 17 Apr 2026 05:01:32 GMT
server: unicorn
```

Responses

Code	Description	Links
200	Successful Response	No links

Media type

application/json

3.2 Reports / Dashboards / Models

Model Used

- Google Gemini 2.5 Flash (LLM)

System Output Structure

The system generates a **Triage Ticket**:

- Original message
- Urgency level
- Intent
- Extracted entities
- Draft reply
- Timestamp

No dashboard currently, but system is ready for:

- Analytics dashboards
- Ticket monitoring systems

3.3 Key Outcomes

- Automated tenant message handling
- Reduced response time (5–15 seconds)
- Accurate classification and extraction
- Human-like AI-generated replies
- Modular and scalable architecture
- Structured and usable output

4. Conclusion

The Real Estate Support Triage Agent successfully demonstrates how AI can automate communication workflows in the real estate domain.

The system:

- Converts unstructured text into structured insights
- Uses a multi-agent pipeline for better accuracy
- Generates professional responses automatically

This project proves that **Agentic AI systems** can significantly reduce manual workload and improve efficiency in real-world applications.

5. Future Scope & Enhancements

The system can be further improved with:

Technical Enhancements

- Database integration (PostgreSQL)
- Async parallel agent execution
- Caching for faster responses

Feature Enhancements

- Real-time chat interface
- Admin dashboard for monitoring
- Ticket tracking system
- Multi-language support

AI Improvements

- Fine-tuned models
- Context memory (conversation history)
- Integration with RAG for knowledge-based queries

Deployment Enhancements

- Cloud deployment (AWS/GCP)
- CI/CD pipeline
- Authentication & authorization